



Cookify

RECIPE RECOMMENDER

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DSAI
DATA SCIENCE
& ARTIFICIAL INTELLIGENCE

Objectives

Build recipe recommender system based on available ingredients

- simplify meal planning
- reduce food waste
- tell users what ingredients they are missing
- combine different models to achieve an optimal recommendation pipeline
- compare bag-of-words vs semantic embedding approaches for recipe similarity

Dataset

- **Source:** Sourced from the Kaggle dataset "Food Ingredients and Recipe Dataset with Images".
- **Dataset Size:** Contains over 13,000 unique, real-world cooking recipes.
- **Key Features Used:** Focuses on recipe titles, raw ingredient lists, cooking instructions, and image pointers (Image_Name).
- **The Main Challenge:** The raw text was full of measurements, weights, and units (like "1 cup", "2 tbsp"), which we had to clean out so our models could focus only on the actual ingredient names.

Development - Five Models

1. **Word2Vec (skip-gram):** Dense word embeddings trained on ingredient corpus, recipe vectors computed by averaging ingredient embeddings, ranked by cosine similarity.
2. **Word2Vec + Lemmatization:** Skip-gram embeddings with preprocessing lemmatization to reduce vocabulary fragmentation and capture semantic relationships between ingredient vectors.
3. **TF-IDF:** Sparse frequency-based vectorization of ingredient strings, capturing ingredient importance across the recipe collection, ranked by cosine similarity.
4. **TF-IDF + Lemmatization:** TF-IDF with lemmatization preprocessing to improve generalization and handle ingredient variations.
5. **SBERT (Sentence-BERT):** Pre-trained transformer-based model encoding full recipe text (title + ingredients) as dense semantic vectors, capturing richer contextual relationships compared to individual embeddings.

Future Work

- **User personalization:** Learn and track user's taste preferences and dietary restrictions
- **Expand recipe list:** Include different, larger datasets
- **Dietary support:** Detect and flag allergies and food intolerances
- **Multi-modal search:** Let users take pictures of their available ingredients
- **Multi-cultural:** Include different languages and sort by traditional cuisines
- **Seasonal availability:** Optimize recommendations based on ingredient seasonality

Conclusion

The project implemented and evaluated five different models, each having different approach. The results showed that lemmatization consistently improves performance across both Word2Vec and TF-IDF models, but is not enough, especially for Word2Vec. SBERT demonstrates strong semantic understanding. Based on this, we conclude that the best recommendation system for this problem is a combination of TF-IDF + Lemmatization with SBERT re-ranking. The results confirm that combining lexical and semantic approaches outperforms any single model. The system also successfully addresses the original objectives of reducing food waste and simplifying meal planning.

Current Limitations

- Model limited only to the recipes included in the dataset.
- Users can choose ingredients exactly as they are in the dataset.
- No user personalization, recommendations identical for all users.
- Cooking methods and equipment requirements not modeled.

Results of five experiments

Model	Precision@5	Recall@5	Coverage
Word2Vec	0.67	0.38	0.48
Word2Vec + Lemmatization	0.80	0.32	0.31
TF-IDF	0.47	0.18	0.22
TF-IDF + Lemmatization	0.73	0.26	0.26
SBERT	0.73	0.39	0.38

Based on the results and known characteristics of these models, the final chosen solution is TF-IDF + Lemmatization with SBERT re-ranking. This combines TF-IDF's efficiency and keyword matching with SBERT's semantic accuracy.

The final model has the following results:

Precision@5	Recall@5	Coverage
1.00	0.87	0.87

*The perfect precision is most likely due to the testing on common ingredients and small evaluation set. Precision on diverse queries is expected to be smaller.

The hybrid model outperforms all individual models, achieving better results across all three metrics, demonstrating the value of combining lexical and semantic strategies.